Revolutionizing Remote Sensing Image Analysis With BESSL-Net: A Boundary-Enhanced Semi-Supervised Learning Network

Zhiyu Yi, Yuebin Wang, *Member, IEEE*, and Liqiang Zhang, *Member, IEEE*

***Abstract*— Deep learning (DL) has become increasingly popular in remote sensing (RS) change detection (CD), leading to the development of massive networks that surpass traditional methods in accuracy and automation. However, the need for enormous amounts of annotated data remains a major concern, and accurate boundary segmentation in RS images is challenging due to their complexity and heterogeneity. Moreover, properly aggregating the bitemporal feature pairs and creating highly discriminative change features are crucial for detection performance. This article proposes a boundary-enhanced semi-supervised network (BESSL-Net) to tackle these issues for CD tasks. The network adopts dual encoders and one decoder architecture for segmentation and incorporates pseudolabeling, contrastive learning (CL), along with a teacher–student scheme to leverage unlabeled data. A boundary extraction module (BEM) is used to conduct boundary segmentation, while a change segmentation feature learning module (SFLM) is used to create discriminative change features in both channel and spatial domains by integrating multilevel features. Three publicly available CD datasets are used to validate the proposed BESSL-Net. Compared with the current state-of-the-art networks, the semi-supervised network demonstrates advanced performance metrics, especially regarding IoU metric of change-class (IoUc), showing improvements ranging from 1% to 9%.**

***Index Terms*— Boundary information, change detection (CD), change features' generation, remote sensing (RS), semi-supervised deep learning (DL).**

I. Introduction

CHANGE detection (CD) in remote sensing (RS) is to extract changes in an object or phenomenon covering the same surface area throughout time based on multitemporal RS images [1]. For various RS applications, including building changes [2], forest monitoring [3], and catastrophe assessment [4], dynamic monitoring of ground objects is essential. The CD technique has therefore been a significant focus of RS [5]. There are limitations to execute CD tasks using satellite and aerial imagery with high spatial resolution, due to several issues including spatial and temporal heterogeneity, infrequent land-cover changes, various scales, and multisources [6]. Despite the benefits of very high-resolution (VHR) imagery in enabling large-scale surface monitoring, these challenges must be addressed to use this technology for CD purposes effectively.

The analysis unit classifies the traditional CD methods into three categories: algebraic methods, transformation methods, and classification methods [7]. The algebraic methods use simple algebra algorithms such as rationing, differencing, and regression to extract change information at the pixel level [8], [9], [10]. The transformation methods, such as the principal component analysis (PCA), the slow feature analysis (SFA), or the tasseled cap transformation, analyze change features and identify change areas [11], [12], [13]. The classification methods such as postclassification comparisons and direct classification rely on classification concepts [14]. In general, the algebraic and transformation methods rely heavily on empirical thresholds, whereas the classification accuracy of each image constrains the classification methods. Hence, due to algorithmic limitations, the traditional CD methods face challenges in producing large-scale and highly accurate CD maps.

Deep learning (DL) techniques have greatly advanced RS and CD tasks. The CD methods based on DL can be categorized into early fusion (EF) and late fusion (LF) networks according to the way to tackle bitemporal images [15]. In EF networks, bitemporal images are first merged as model inputs by concatenation or difference [16]. LF networks, such as Siamese networks, perform separate feature representation learning on bitemporal images using dual encoders [15], which have improved performance than EF networks in CD tasks [17]. However, accurate results based on DL networks require a huge number of high-quality labels, which can be expensive and time-consuming to be manually collected due to the need for semantic comparison of two images.

To address the limitation of labeled data in semantic segmentation, semi-supervised learning has been used to extract knowledge from unlabeled data, frequently using pseudolabeling consistency regularization. Motivated by

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entropy minimization [18], pseudolabeling assigns synthetic labels to unlabeled data while maintaining reliable predictions from an extant model [19]. The mean teacher method [20] has recently been applied for consistency regularization with a teacher–student scheme, in which weights of the teacher network are optimized with exponential moving average (EMA) to provide more reliable predictions. As a data-oriented technique, semi-supervised methods for semantic segmentation almost exclusively deal with natural RGB images such as ImageNet [21], which are different from RS imagery with large and complex objects and scenes. Moreover, semi-supervised CD (SS-CD) methods disregard the importance of boundary information to improve region accuracy. The rise in intraobject variability and the complexity of boundaries have posed significant obstacles to successful segment boundaries due to the continual enhancement of the spatial resolution of RS imagery.

In this work, a novel SS-CD network called boundary-enhanced semi-supervised network (BESSL-Net) for VHR RS imagery is proposed that not only accurately identifies change areas with limited labeled data but also considers the boundary information while extracting change features more reasonably. Based on a novel semi-supervised segmentation module [22], BESSL-Net is built on dual encoders and one decoder architecture and incorporates pseudolabeling, contrastive learning (CL), along with a teacher–student scheme to leverage unlabeled data. Moreover, a boundary extraction module (BEM) is proposed between the dual encoders and the decoder to perform the boundary detection without additional computation or a new loss function. We also introduce a change segmentation feature learning module (SFLM) that integrates multilevel features from dual encoders and provides identifiable change features. The proposed BESSL-Net is evaluated on three publicly available CD datasets. This article makes several significant contributions, including the below.

1) A novel network, BESSL-Net, is proposed for SS-CD tasks, which combines BEM and SFLM to enhance CD performance by emphasizing boundary detection and distinguishing change feature extraction. The BESSL-Net successfully addresses the challenge of limited labeled data while also taking boundary information into account.
2) Without the requirement for extra computation or new loss functions, the BEM is used to explore the boundary information of primitive and semantic feature vectors. Rather than just combining bitemporal feature pairs, the SFLM generates discriminative change features using different combinations of the primitive and semantic feature vectors, incorporating an attention mechanism.
3) The superiority of BESSL-Net is highlighted in the validation on three public datasets, particularly in achieving remarkable performance improvements with minimal labeled data (5%).

II. Related Work

This section offers a concise summary of related methods for CD tasks, with a particular emphasis on SS-CD and boundary-enhanced semantic segmentation.

A. Semi-Supervised CD

Recent researchers have introduced semi-supervised approaches to CD tasks to address label inefficiency due to the incredible ability of semi-supervised learning techniques to extract unlabeled information and the low dependency on labels. Yuan et al. [23] used Laplacian regularization metric learning to incorporate information from unlabeled images. A coarse-to-fine network [24] is presented for SS-CD tasks with metric learning and neighborhood label information. But these SS-CD methods are conducted on individual image pairs rather than on a large number of unlabeled images.

Generative adversarial networks (GANs) have also been applied in CD tasks to overcome label deficiency. A type of GAN called generative discriminatory classified network (GDCN) [25] has been used based on multispectral imagery, which generated synthetic data using random noise. But generating synthetic data using only random noise has limitations for VHR images with complex scenes. Peng et al. [26] developed a semi-supervised network that exploited the information of unlabeled images to continuously regularize discriminators and boost the model performance. Apart from GAN, consistency regularization and pseudolabeling have also been introduced into semi-supervised RS semantic segmentation and CD tasks. PanPaste [27] is proposed for RS semantic segmentation, which simultaneously uses consistency regularization, pseudolabeling, and thresholds to gradually enhance the model performance. Bandara and Patel [28] developed a consistency loss with random perturbations on the deep feature difference maps to restrict the prediction maps of unlabeled bitemporal image pairs. Revisiting consistency regularization boosts the robustness, generalization, and feature extraction capabilities using complex perturbations.

Recently, CL has gained significant attention. The core idea of CL is gathering positive data pairs and separating negative ones in the feature space [29]. As for semi-supervised RS semantic segmentation and CD, CL techniques have been mainly used in the pretraining stage. A novel network based on a nonsmooth nonnegative matrix factorization network [30] is proposed for synthetic aperture radar image CD, where all the decomposed matrices are pretrained layer by layer. Chen et al. [31] explored the importance of semantic information during the pretraining stage by collecting large-scale images from the RS community for pretraining, rather than using the ImageNet dataset. In addition to pretraining, researchers have also proposed end-to-end networks based on CL techniques for CD tasks. A method using reliable CL (RCL) [32] is developed with the generation of a contrastive loss for change areas and it selects reliable pseudolabels based on the uncertainty in unlabeled data.

B. Boundary-Enhanced Semantic Segmentation

Researchers have directed their attention toward improving both the accuracy of semantic segmentation and the capacity to extract boundary information due to the significance of boundary information in semantic segmentation. Early investigations extracted primitive boundary information with boundary detection operators such as Sobel [33] and Laplace

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of Gaussian. To refine the segmentation results, researchers also conducted boundary extraction as a postprocessing step using a boundary neural field [34], adaptive affinity fields [35], or a random walk [36].

Recently, there has been a growing trend to create a separate subnetwork for boundary detection alongside the semantic segmentation architecture to accurately detect ambiguous pixels. A two-stream CNN [37] is proposed to explicitly wire shape information as an independent branch alongside the semantic segmentation steam. A novel framework [38] is proposed to jointly tackle inner object consistency and boundary delineation by individually analyzing the body and boundary features. For RS semantic segmentation, a semi-supervised network [39] assigns higher weights to features with abundant boundary messages using an attention module.

Moreover, researchers have concentrated on designing new loss functions to handle the boundary pixels. EaNet [40] was proposed for urban semantic segmentation and presented a novel loss to assist the network in refining boundary messages both at the pixel and image scales. Another innovative approach is developing a boundary-aware loss [41], which captures the extent of transformations between the predicted and actual boundaries.

III. Proposed Model

This section elaborates on the specifics of our proposed network, BESSL-Net. We start with the motivation for BESSL-Net in Section III-A, followed by the overall structure of our model in Section III-B. Then, the BEM (Section III-C) and the change SFLM (Section III-D) are demonstrated in detail. Finally, the optimization of BESSL-Net is described in Section III-E.

A. Motivation

Insufficient labeling remains a persistent issue in RS CD tasks. One promising solution to this issue is semi-supervised learning, which explores information from unlabeled data to supplement labeled data. Moreover, RS images tend to be larger and contain more objects than natural images, resulting in a higher fraction of boundary pixels and more boundary blur. As the spatial resolution increases, it becomes more difficult to reliably segment boundary pixels for CD tasks due to the variety in object appearance and complexity of object boundaries. In addition, although Siamese networks are effective for CD tasks, researchers frequently only concatenate or subtract temporal image features obtained from dual encoders, which overlooks crucial features for discriminating change areas and harms model performance. Therefore, for optimal detection performance, it is imperative to focus on properly aggregating the bitemporal feature pairs and creating highly discriminative change features.

To address these challenges, this article presents a novel CD model built upon a semi-supervised segmentation backbone. Specifically, we use pseudolabeling, CL, and a teacher–student scheme to explore unlabeled information. Then, we suggest a BEM to conduct the boundary segmentation. In addition, we establish a change SFLM to integrate primitive and semantic features from the dual encoders and generate identifiable change features at channel and spatial levels.

B. Overall Structure

The training data are split into two sections: labeled data $x_l = \{i_{l1}, i_{l2}, y_l\}$ ($\mathbf{y}_l$ denotes the corresponding ground-truth labels) and unlabeled data $x_u = \{i_{u1}, i_{u2}\}$. $i \in \mathbb{R}^{H\times W\times C}$ is VHR RS image pair, I_1 and I_2 ($H \times W$ denotes the quantity of pixels and C represents the amount of bands).

The BESSL-Net architecture, as illustrated in Fig. 1, uses a semi-supervised Siamese segmentation backbone, a variation in DeepLabV3+ [42] with two weight-sharing encoders and one decoder. To leverage the potential of x_u, a teacher–student scheme is used, where a teacher model generates reliable pseudolabels from x_u. A classwise feature bank is also designed based on the teacher model to store high-quality labeled features from different classes and enforce consistent pixelwise feature representations within the same class across the entire dataset. The student model is optimized with a supervised loss from x_l, a pseudolabeling loss from x_u, along with a CL loss on the fundamentals of the feature Bank. In addition, to boost the generation of discriminative change features, we propose two modules connecting the dual encoders with the decoder: the BEM and the change SFLM. The BEM is used to serve boundary detection, while the SFLM is applied for the channelwise and spatialwise fusion of the discriminative change features. Both the modules are relatively simple methods without the addition of excessive parameters and computations.

C. Boundary Extraction Module

Inspired by the BGNet [43], we propose a BEM (Fig. 2) to effectively extract boundary pixels in RS images without more loss functions. This module comprises three submodules: an edge-aware module, two edge-guidance feature modules, and a context aggregation module. In general, primitive features include a wide range of boundary details along with extensive nonobject boundaries. In contrast, semantic features provide a wealth of semantic and geographical information that can compensate for the limitations of primitive features. Hence, a combination of both types of features is commonly used to extract boundary information from RS images. The dual encoders generate feature sets that are performed separate concatenations to produce primitive features (F_l) and semantic features (F_h), both of which are used as inputs for BEM.

More specifically, the edge-aware module incorporates a few straightforward convolutional layers to extract boundaries. Abundant boundary information from F_l and extensive global information from F_h after being upsampling are combined and input into the module to generate a boundary-enhanced feature F_a, formulated as follows:

$$F_a = \sigma\left(f^1\left(f^3\left(f^3\left[U\left(f^1(F_h) + f^1(F_l)\right)\right]\right)\right)\right) \tag{1}$$

where σ and U denote a sigmoid function and upsampling, respectively. f^i is a convolutional layer with a kernel size of i, while $+$ denotes concatenation.

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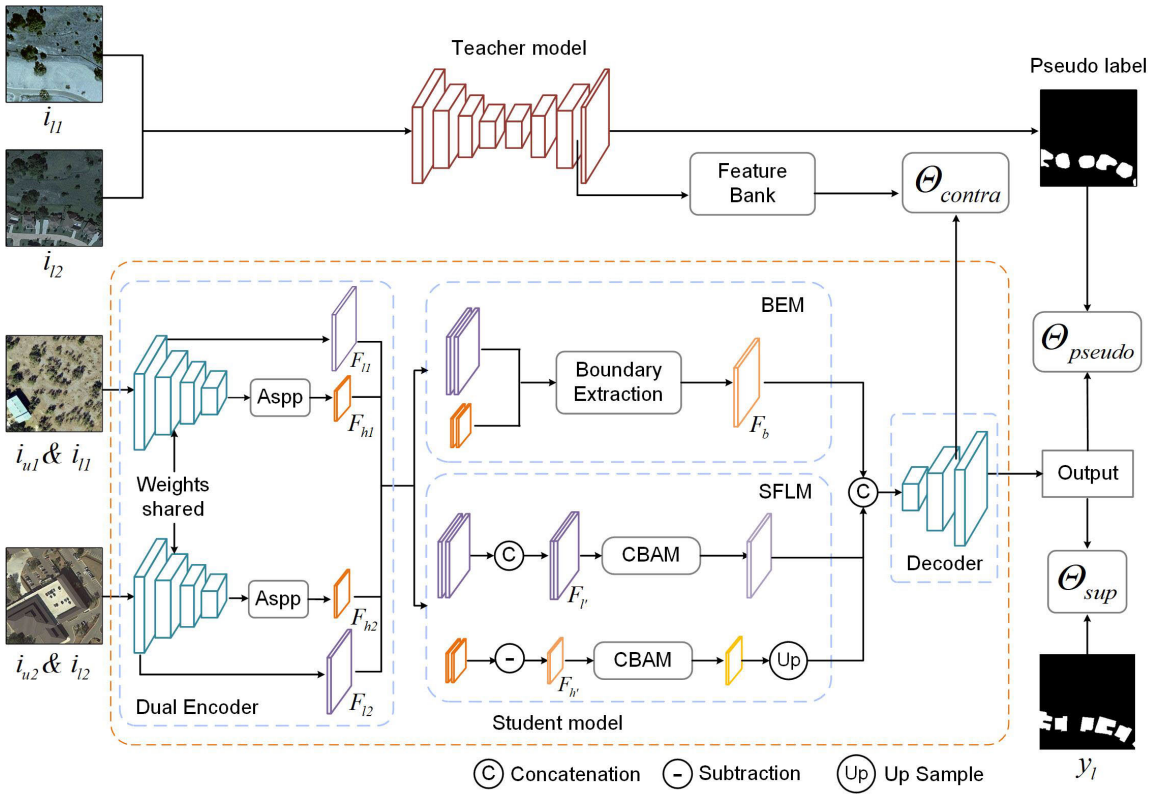


Fig. 1. Flowchart of BESSL-Net.

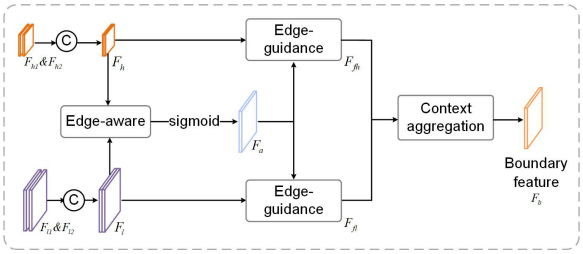


Fig. 2. Structure of BEM, which contains three main components: an edge-aware module, two edge-guidance feature modules, and a context aggregation module.

The edge-guidance modules are applied to create a more cohesive and robust representation by integrating important information from multiple channels, incorporating two parameter-free mathematical operations, multiplication, addition, and channel attention mechanisms. Initial fused features F_l^a and F_h^a are created based on the boundary-enhanced feature F_a and either the initial feature F_l or the feature F_h, defined as (2). Next, the features F_l^a and F_h^a are processed with local cross-attention interaction to produce the fused features F_{fh} and F_{fl}, formulated as (3)

$$\begin{aligned} F_l^a &= f^3((F_l \otimes F_a) \oplus F_l) \\ F_h^a &= f^3((F_h \otimes D(F_a) \oplus F_h) \end{aligned} \tag{2}$$

where D denotes downsampling. $\otimes$ and $\oplus$ are elementwise multiplication and addition, respectively

$$F_{fi} = f^1\left(\sigma\left(f_{lD}^j\left(\text{AvgPool}\left(F_i^a\right)\right)\right) \otimes F_i^a\right) \tag{3}$$

where $f_{1\text{-D}}^j$ is the 1-D convolution layer with kernel size j. And j can be defined as $j = |(1 + \log_2(C))/2|_{\text{odd}}$, $|*|_{\text{odd}}$ denotes the closest odd number, and C denotes the channels of F_i^a (i denotes l or h).

Finally, the context aggregation module performs cross-scale interactions to fuse multilevel features and enhance the representation of boundary information across different branches. First, an initial aggregated feature F_c is created based on the fused features F_{fl} and F_{fh} after being unsampled. The feature F_c is then divided into four feature maps F_{ci} to enable cross-scale interactions, resulting in the generation of the boundary feature F_b, which is defined using the following equation:

$$F_b = f^3\left(F_c \oplus f^1\left(F'_{c1} + F'_{c2} + F'_{c3} + F'_{c4}\right)\right) \tag{4}$$

where F_c and F_{ci} are defined as

$$\begin{aligned} F_c &= f^1\left(F_{fl} + U\left(F_{fh}\right)\right) \\ F'_{ci} &= f_i^3\left(F'_{ci-1} \oplus F_{ci} \oplus F_{ci+1}\right) \end{aligned} \tag{5}$$

where f_i^3 denotes the 3×3 atrous convolutional layer with a dilation rate of $i \in \{1, 2, 3, 4\}$. For $i = 1$, there is only F_{c1} and F_{c2}; for $i = 4$, there is only F'_{c3} and F_{c4}.

D. Segmentation Feature Learning Module

It is essential for detection performance that there is a proper aggregation of the bitemporal feature pairs and the creation of highly discriminative change features. Semantic information cannot be adequately conveyed by aggregating bitemporal

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features based on elementwise addition or channel concatenation [44]. To exploit the ample spatial information from primitive features, as well as abundant semantic messages in semantic features, we conduct different combinations for these features generated by the dual encoders (Fig. 1). Specifically, the semantic features of Image 2 are subtracted from those of Image 1 to generate the final semantic features F_h', whereas the primitive feature F_l' is produced through concatenating two primitive features, which is denoted as follows:

$$\begin{aligned} F_l' &= F_{l1} + F_{l2} \\ F_h' &= F_{h2} - F_{h1} \end{aligned} \tag{6}$$

where $+$ denotes concatenation, and $-$ represents elementwise subtraction.

Furthermore, for the purpose of enhancing the distinctiveness of these features, both F_l' and F_h' undergo processing with a convolutional block attention module (CBAM) [45]. This attention module is lightweight without excessive memory and computing overhead and can concentrate on both the channel and spatial dimensions to enhance valuable information in the retrieved features across multiple dimensions. CBAM infers a 1-D channel attention feature followed by a 2-D spatial attention feature. This can be computed using the following equation:

$$\begin{aligned} F_1' &= \sigma\left(\mathrm{MLP}^{\gamma}\left(\mathrm{AvgP}\left(F'\right)\right) + \mathrm{MLP}^{\gamma}\left(\mathrm{MaxP}\left(F'\right)\right)\right) \otimes F' \\ F_2' &= \sigma\left(f^{7}\left(\mathrm{AvgP}\left(F_1'\right) + \mathrm{MaxP}\left(F_1'\right)\right)\right) \otimes F_1' \end{aligned} \tag{7}$$

where F_2' is the CBAM output. σ and γ are the sigmoid function and the reduction ratio, respectively. AvgP and MaxP denote the average pooling and maximum pooling operations, respectively.

Finally, to generate change feature maps, the boundary features F_b obtained via the BEM are concatenated with primitive features processed with SFLM and semantic features obtained from SFLM by upsampling the same size as the primitive feature.

E. Optimization of BESSL-Net

The optimization of the student model in the teacher–student scheme involves minimizing a loss function that is defined in the following equation:

$$\Theta = \Theta_{\mathrm{sup}} + \lambda_1 \Theta_{\mathrm{pseudo}} + \lambda_2 \Theta_{\mathrm{contra}} \tag{8}$$

where λ_1 and λ_2 are hyperparameters for loss weighting. There are three loss terms in (8). Θ_{sup} is a weight cross-entropy loss on x_l using supervised learning (9). Θ_{pseudo} is the pseudolabeling loss (10). Θ_{contra} is used to conduct CL (11)

$$\Theta_{\mathrm{sup}} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{c=1}^{C} y_l^{(n,c)} \log\left(\hat{y}_l^{(a,n,c)}\right)\rho^c \eta^n \tag{9}$$

where y_l and $\hat{y}_l$ represent the actual labels and the prediction obtained from the student model based on x_l, respectively. N denotes the total number of pixels, while C is 2, which denotes the classification number. ρ^c and η^n denote per-class and per-pixel weight, respectively

$$\Theta_{\mathrm{pseudo}} = -\frac{1}{A}\frac{1}{N}\sum_{a=1}^{A}\sum_{n=1}^{N}\sum_{c=1}^{C} \hat{y}_u^{(n,c)} \log\left(f_\beta(x_u)^{(a,n,c)}\right)\rho^c \eta^n \tag{10}$$

where $f_\beta(x_u)$ is the prediction created by the student model of x_u, and A denotes the number of augmentations applied to images x_u

$$\Theta_{\mathrm{contra}} = \frac{1}{C}\frac{1}{N_{p_c}}\frac{1}{N_{z_c'}}\sum_{c=1}^{c}\sum_{p_c\in P_c}\sum_{z_c'\in Z_{c'}} W_{p_c} W_{z_c'}\left(1-\cos\left(p_c, z_c'\right)\right) \tag{11}$$

where p_c and z_c', respectively, denote the feature vector of x_u and the corresponding high-quality feature vector in feature bank based on x_l. w_{p_c} and $w_{z_c'}$ are the weights of p_c and z_c' acquired from class-specific attention modules. The similarity between p_c and z_c' is maximized by measuring the weight distance with cosine similarity, computed as $\cos(p_c, z_c') = (\langle p_c, z_c'\rangle / \|p_c\|_2 \cdot \|z_c'\|_2)$.

What is more, rather than using gradient descent, the weights α of the teacher model are optimized through the EMA of weights β of the student network with a decay rate $\gamma \in [0, 1]$, formulated as follows:

$$\alpha = \gamma\alpha + (1-\gamma)\beta. \tag{12}$$

TABLE I
BRIEF INTRODUCTION OF THE EXPERIMENTAL DATASETS

Dataset	Bands	Image pairs	Resolution(m)	Image size
LEVIR-CD	3	637	0.5	1024*1024
WHU-CD	3	1	0.3	32207*15354
SYSU-CD	3	20000	0.5	256*256

IV. EXPERIMENTS AND DISCUSSION

To access the validation of BESSL-Net, we perform experiments on three challenging open-source datasets. This section begins with a detailed description of experimental setups, including specifics of three datasets and implementation. We then briefly demonstrate several SOTA semi-supervised approaches for comparison. Afterward, the experimental results are analyzed in detail.

A. Experimental Setup

1) Datasets: This section is devoted to the presentations of experimental datasets, with a succinct summary of the datasets provided in Table I.

a) LEVIR-CD Dataset: This is a commonly used dataset [46] for large-scale CD that includes various kinds of buildings, as shown in Fig. 3(a). It comprises 637 VHR image pairs, each having a resolution of 0.5 m and a size of 1024 × 1024. The dataset is divided into training, validation, and testing sets with a ratio of 7:1:2 based on the default split defined in [45]. To reduce computational costs, each image is divided into 256 × 256 nonoverlapping patches. We also crop each image into 512 × 512 nonoverlapping patches to examine the effect of different training patch sizes.

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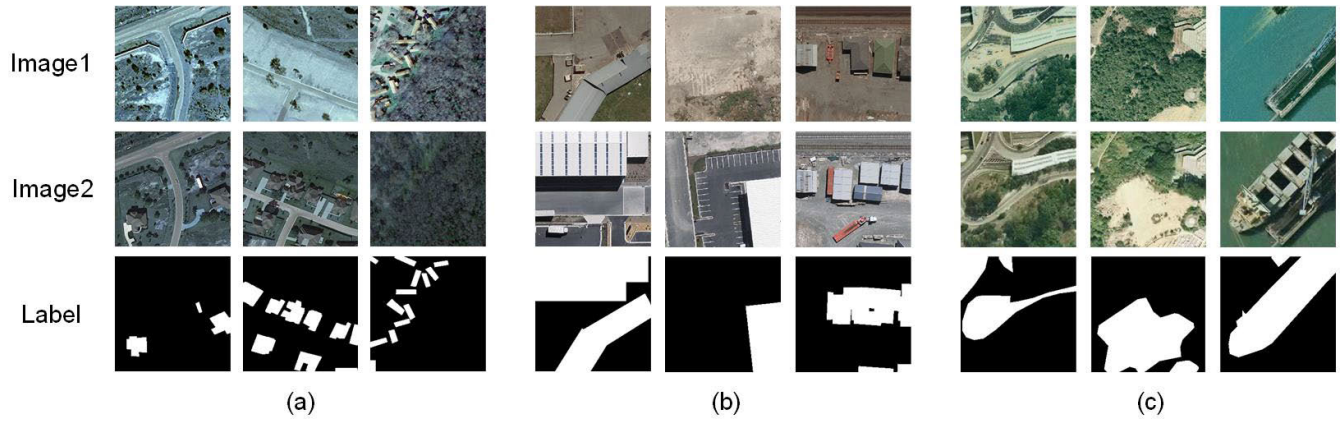


Fig. 3. Example samples (256 × 256) from the experimental datasets. (a) Samples from the LEVIR-CD. (b) Samples from the WHU-CD. (c) Samples from the SYSU-CD.

b) WHU-CD Dataset: This dataset [47] contains building changes in Christchurch, New Zealand, using a pair of aerial images captured in 2012 and 2016 [Fig. 3(b)]. These images are 0.3-m spatial resolution with a large size of 32 207 × 15 354. Similar to the LEVIR-CD dataset, each image is sliced into 256 × 256 and 512 × 512 segments without overlap. In addition, since there is no normal partition, images are randomly allocated to training, validation, and test sets at random in a ratio of 6:2:2.

c) SYSU-CD Dataset: It is a brand-new open-source CD dataset [48]. In contrast to the two datasets, SYSU-CD contains information on common changes in urban and suburban areas and changes in vegetation and sea construction, resulting in more irregular boundaries of its change objects [Fig. 3(c)]. This dataset is made up of 20 000 pairs of 256 × 256 aerial image patches, each with a spatial resolution of 0.5 m. The default spillover ratio of 6:2:2 is adopted for training, validation, and testing sets.

2) Implementation Details:

a) Data Preprocessing: Training/validation/testing sets are created as described in Section IV-A1. Following division, we find that the training set contains a sizable number of unchanged image pairs. To assist the model in learning the features of the changing area more efficiently, we eliminate these image pairs with only the nonchange category from the training set. Finally, there are 3167/1024/2048 image pairs for the respective training, validation, and testing sets in the LEVIR-CD. The WHU-CD has 1162, 1536, and 1536 image pairs in training, validation, and testing. In all, 11 782 training, 4000 validation, and 4000 testing sets make up the SYSU-CD. Before being fed into the network, each patch is normalized using mean and standard deviation.

Following [22], two different data augmentation strategies are applied for labeled and unlabeled images, a weak one for x_l and a strong one for x_u. What is more, parameter A in (10) is set to 2 as the number of augmentations of each sample.

b) Evaluation Metrics: Following [28], [32], the generalization capacity of our CD model is measured using intersection over union (IoU) as the primary performance metric to determine the best semi-supervised model. The overall accuracy (OA), which measures the percentage of correctly predicted pixels, is also applied in conjunction with the IoU metric of change-class (IoUc). The values of IoUc and OA range from 0 to 1, with larger values indicating better performance.

c) Training Details: Owing to the limited available memory on the GPU during training, the minibatch is set to 16 for 256 × 256 in all the datasets and four for experiments based on 512 × 512 images for the LEVIR-CD and WHU-CD. The hyperparameters are configured with some minor alterations to [22]. All the models are trained using the SGD optimizer with momentum 0.9 for 150 k iterations. With a poly learning rate schedule, the learning rate for the LEVIR-CD and SYSU-CD is set to 0.01, whereas it is set to 0.04 for the WHU-CD. What is more, γ in (5) is set from 0.995 to 1 for training. The values for λ_1 and λ_2 are set to 1 and 0.1, respectively (8). It should be noted that λ_1 and λ_2 are set to 0 for the first 2 k iterations to generate a high-quality feature bank. ρ^c in (9) and (10) is set to 1 due to the insignificant benefit of class balancing. The checkpoint exhibiting the highest IoUc on the validation set is also applied for testing.

B. Comparative Methods

To assess the effectiveness of BESSL-Net, the following SOTA semi-supervised semantic segmentation and CD models are applied for comparison, including AdvNet [49], S4GAN [50], SemiCDNet [26], SS-CD [28], and RCCD [32]. The specifics are given as follows.

1) *AdvNet:* AdvNet is a semi-supervised semantic segmentation model using entropy loss and adversarial loss. The model introduces a fully convolutional discriminator to assist with the consistency of the feature distribution between labeled predictions and actual masks. In addition, AdvNet incorporates a self-training technique that uses the high-confidence areas of unlabeled predictions.
2) *S4GAN:* S4GAN is based on pixelwise class labels. To improve the training of GANs, a novel loss function is used to match features and minimize the difference between the predictions and the actual maps. The generator and discriminator are additionally balanced using a self-training loss.

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TABLE II
COMPARISON RESULTS ON THE LEVIR-CD DATASET WITH DIFFERENT PERCENTAGES OF LABELED DATA

Method	Labeled Ratio							
	5%		10%		20%		40%	
	IoUc	OA	IoUc	OA	IoUc	OA	IoUc	OA
AdvNet	0.2801	0.9405	0.4664	0.9666	0.4808	0.9663	0.5215	0.9685
S4GAN	0.5183	0.9658	0.6003	0.975	0.6528	0.9781	0.6814	0.9807
SemiCDNet	0.6515	0.9796	0.712	0.9834	0.7232	0.9846	0.7526	0.9862
SS-CD	0.6788	0.9807	0.7298	0.9852	0.7634	0.9867	0.7626	0.9867
RCCD	0.7265	0.9839	0.7236	0.9832	0.7468	0.9849	0.7595	0.9859
BESSL-Net	**0.8029**	**0.989**	**0.8114**	**0.9895**	**0.8143**	**0.9896**	**0.8183**	**0.9899**

The best results are highlighted in bold.

3) *SemiCDNet:* SemiCDNet is a convolutional model built on a GAN for CD tasks. Continuous regularization is performed with the unlabeled images to enhance the generalization capability of the generator.
4) *SS-CD:* SS-CD model is based on revisiting consistency regularization. To improve the stability of the model, noise is added to high-feature maps and compelled to match the predictions without noise.
5) *RCCD:* RCCD is the current SOTA method for SS-CD tasks. A novel contrastive loss and reliable pseudolabels are applied to use the messages in unlabeled data effectively.

We retain default hyperparameters for all the SOTA methods. The sampling ratio is set to 5%, 10%, 20%, and 40% for all the three datasets. To ensure fair comparisons, all the experiments use the same data splits for labeled and unlabeled images.

C. Comparison Results

1) LEVIR-CD: On the LEVIR-CD, we compare the performance of BESSL-Net with various semi-supervised methods in this section. Table II presents the comparison results for 5%, 10%, 20%, and 40% labeled images in the training dataset using two evaluation metrics, IoUc and OA.

In general, semi-supervised methods tend to achieve higher accuracy with more labeled data. However, there are a few outliers, as indicated in Table II. For instance, the results of RCCD using 10% labeled images are better than those using 20% labeled images. This could be due to the effect of our data preprocessing procedure on the results or because the method is designed to achieve high performance with less labeled data. In addition, it is noted that CD methods using RS images clearly outperform other semi-supervised segmentation methods proposed for natural images (i.e., AdvNet and S4GAN). This suggests that the features of RS images differ substantially from those of natural images and that SS-CD merits further exploration. Most notably, our BESSL-Net significantly outperforms the most recent SS-CD methods, achieving the highest values of both the metrics with at least a 6% increase in the IoU metric. This supports the effectiveness of our BEESL-Net, which performs better in extracting changed areas.

In Fig. 4, we present typical CD results on the test set with different label ratios for visual comparison. As observed, the amount of labeled images considerably impacts the identification ability of a model in semi-supervised learning. The SS-CD methods deliver respectable results even with a few labeled images, in contrast to the two segmentation methods, which exhibit relatively poor results due to the complex and diverse types of surface objects in RS images. Furthermore, our BESSL-Net achieves similar results to the ground truth, even with only 5% labeled images. The boundary identification of our model is more precise regardless of the proportion of labeled images. Therefore, the visual results are in line with the quantitative findings, indicating that BESSL-Net is effective in detecting changed areas and boundaries.

2) WHU-CD: This section provides the comparative results of BESSL-Net against several SOTA methods on the WHU-CD. Table III illustrates the quantitative results for IoUc and OA.

Table II demonstrates that our BESSL-Net outperforms all other semi-supervised methods with an impressive advantage on the WHU-CD dataset across four labeled ratios. BESSL-Net achieves the highest IoUc and OA. While having little impact on OA, the labeled ratio has a considerable influence on IoUc. This shows that the model becomes increasingly sensitive to changed samples as the number of labeled images grows. Moreover, the benefit of BESSL-Net is more apparent when there are fewer labeled images. The IoUc is only 1% higher than the recent SS-CD method, RCCD, with 40% labeled images, whereas the increase rises to 5% with only 5% labeled data. Hence, our model can better exploit the information in unlabeled data and overcome the insufficiency of labeled data in current applications.

Fig. 5 provides a clear and intuitive visualization of the performance of various methods on the WHU-CD. As illustrated in Fig. 5(a), the comparative semi-supervised methods have varying levels of mistake detection, and the precision of boundary detection is relatively low, as evidenced by the rough boundary in the bottom left of the results. On the other hand, our method achieves satisfactory accuracy in boundary detection without errors or omissions. In addition, Fig. 5(b) demonstrates that our BESSL-Net outperforms other semi-supervised approaches in detecting both the change areas and boundaries.

3) SYSU-CD: The section demonstrates the comparison performance of BESSL-Net with other comparison methods

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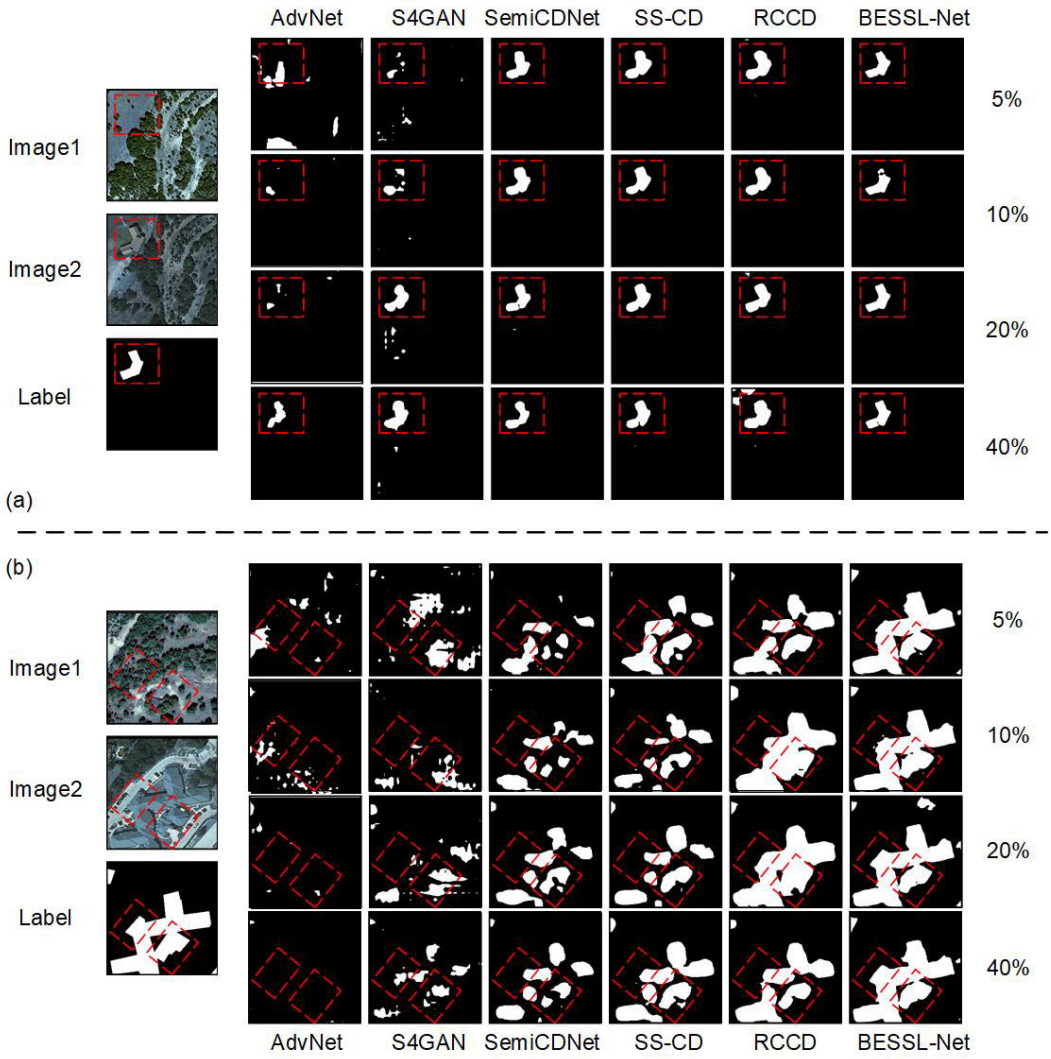


Fig. 4. Visualization comparisons of various networks on the LEVIR-CD testing set using different label data percentages. (a) and (b) Different image pairs.

TABLE III
COMPARISON RESULTS ON THE WHU-CD DATASET WITH DIFFERENT PERCENTAGES OF LABELED DATA

Method	Labeled Ratio							
	5%		10%		20%		40%	
	IoUc	OA	IoUc	OA	IoUc	OA	IoUc	OA
AdvNet	0.2737	0.9589	0.3681	0.9522	0.4618	0.9697	0.4972	0.9667
S4GAN	0.4379	0.962	0.5066	0.9681	0.5965	0.9778	0.6316	0.98
SemiCDNet	0.5405	0.9744	0.6372	0.9813	0.6885	0.9845	0.761	0.9884
SS-CD	0.641	0.9827	0.6714	0.9835	0.7289	0.9868	0.7772	0.9897
RCCD	0.6827	0.9849	0.7114	0.9863	0.7488	0.9879	0.8016	0.9906
BESSL-Net	**0.7393**	**0.9877**	**0.7807**	**0.9899**	**0.7902**	**0.9905**	**0.8105**	**0.9912**

The best results are highlighted in bold.

on the SYSU-CD. Four proportions of labeled images in the training set are also used for comparison of all the methods, with evaluation results presented in Table IV.

According to Table IV, the proposed BESSL-Net outperforms all the comparisons on the SYSU-CD with the highest overall OA and the IoUc. The benefits of BESSL-Net are particularly evident when dealing with a small number of labeled images. With only 10% labeled images, BESSL-Net outperforms the best-performing comparison, SS-CD, by 2% in IoUc and 1% in OA. This also demonstrates that our network can better explore the information of unlabeled images and overcome the insufficient availability of labeled

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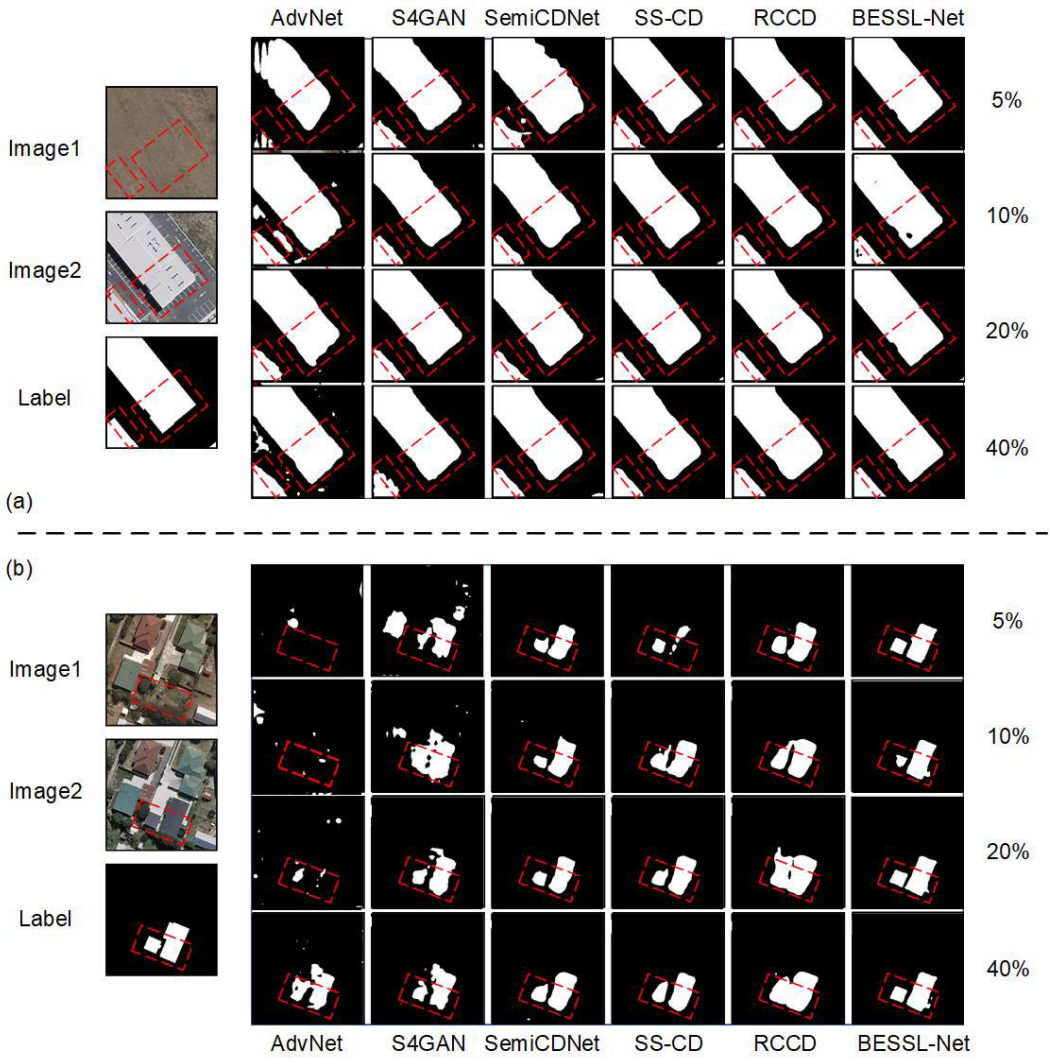


Fig. 5. Visualization comparisons of various networks on the WHU-CD testing set using different label data percentages. (a) and (b) Different image pairs.

TABLE IV
COMPARISON RESULTS ON THE SYSU-CD DATASET WITH DIFFERENT PERCENTAGES OF LABELED DATA

Method	Labeled Ratio							
	5%		10%		20%		40%	
	IoUc	OA	IoUc	OA	IoUc	OA	IoUc	OA
AdvNet	0.4348	0.7575	0.477	0.7953	0.45	0.8064	0.4547	0.8263
S4GAN	0.6059	0.8895	0.6242	0.8948	0.6381	0.899	0.6467	0.9026
SemiCDNet	0.6057	0.8916	0.6388	0.8998	0.6529	0.9057	0.6755	0.9121
SS-CD	0.5268	0.8606	0.6417	0.9001	0.6589	0.9086	0.6604	0.9047
RCCD	0.6353	0.9028	0.6278	0.8974	0.641	0.9031	0.6606	0.9048
BESSL-Net	**0.6451**	**0.9078**	**0.6687**	**0.9143**	**0.6774**	**0.912**	**0.6828**	**0.9147**

The best results are highlighted in bold.

data in current applications. In contrast to the results on the other two datasets, the SS-CD methods do not significantly outperform the segmentation methods proposed for natural images. The IoUc achieved using all the semi-supervised methods is centered around 60%. We hypothesize that this is due to the diverse types of ground objects in the SYSU-CD dataset, such as vegetation, which are strongly affected by seasonal variations. These factors may present challenges to precisely identifying changed areas.

A visual comparison of all semi-supervised networks on the SYSU-CD is depicted in Fig. 6. The results of the comparative methods exhibit several errors and omissions. In contrast, our

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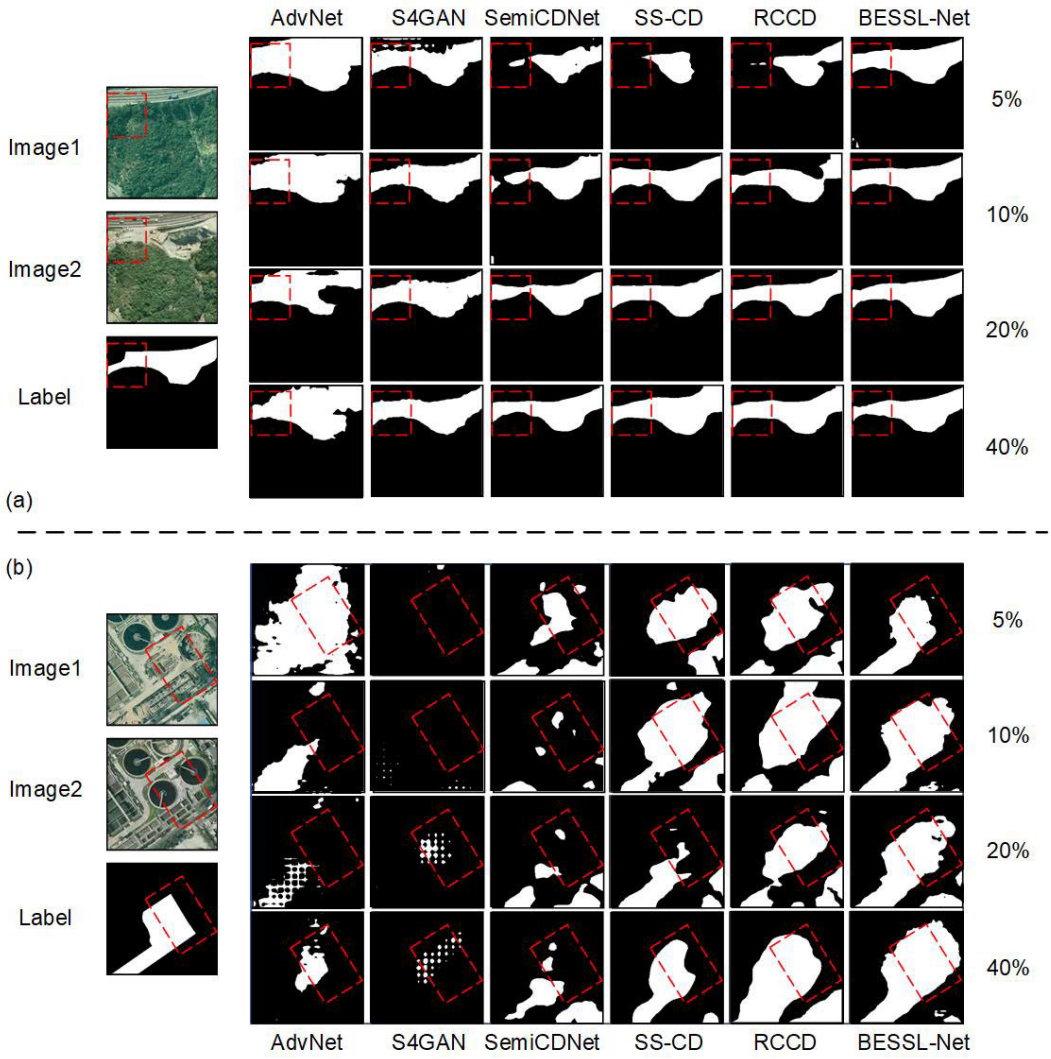


Fig. 6. Visualization comparisons of various networks on the SYSU-CD testing set using different label data percentages. (a) and (b) Different image pairs.

BESSL-Net gains the most accurate detection performance and produces more consistent results with the ground truth [Fig. 6(a)]. However, according to the example in Fig. 6(b), all the semi-supervised methods perform poorly, with numerous errors and omissions. This suggests that conducting CD tasks based on the SYSU-CD dataset is challenging. In addition, our method performs well with only 5% labeled data, but with an increase in labeled images, many regions are incorrectly identified as changed areas. The original image pairs reveal that various factors, such as coverage and the presence of water, directly interfere with the identification accuracy of the model.

Furthermore, the accuracy of all the semi-supervised methods on the SYUS-CD dataset is much worse than that of the other two datasets. The SYSU-CD dataset contains more complex surface objects and is greatly impacted by seasonal variations compared with building datasets such as the LEVIR-CD and WHU-CD datasets. As a result, RS CD using the SYSU-CD dataset still presents several challenges that require attention. Further exploration of the SYSU-CD is crucial for developing more accurate CD methods in complex and variable environments.

D. Effectiveness Analysis

We discuss the impact of the innovations suggested in this article on BESSL-Net in this section. Specifically, ablation experiments are conducted on BESSL-Net to evaluate the BEM and the SFLM. The optimization of BESSL-Net using different loss functions is then analyzed. In addition, we assess the validity of data preprocessing techniques used on various datasets, including removing all the images with only one classification in the training set and adjusting the crop size in the LEVIR-CD and WHU-CD. The validation is also evaluated using two evaluation metrics: IoUc and OA.

1) BEM and SFLM: Starting from a semi-supervised learning model [22] with dual encoders and one decoder backbone, the proposed method suggests BEM and SFLM for accurate discrimination of boundary pixels and change features. To assess the reliability of both BEM and SFLM, we perform ablation experiments on the baseline and its variants on the

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TABLE V
ABLATION STUDY ON THE THREE DATASETS WITH 10% LABELED DATA IN TRAINING

Method	LEVIR-CD		WHU-CD		SYSU-CD	
	IoUc	OA	IoUc	OA	IoUc	OA
Base	0.7861	0.9885	0.7426	0.9878	0.6239	0.8974
Base+BEM	0.8103	0.9894	0.7695	0.9896	0.6501	0.9095
Base+SFLM	0.7868	0.9877	0.7446	0.988	0.5995	0.8927
BESSL-Net	**0.8114**	**0.9895**	**0.7807**	**0.9899**	**0.6687**	**0.9143**

The best results are highlighted in bold.

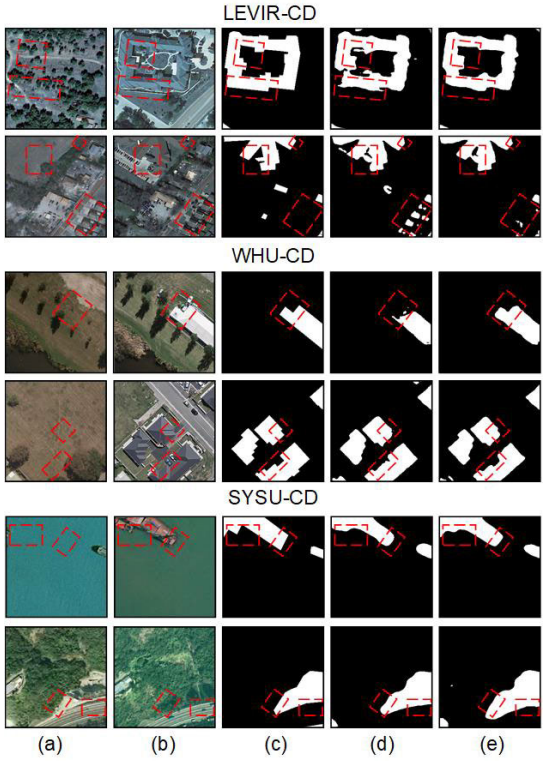


Fig. 7. Visualization results of BEM. (a) Image 1. (b) Image 2. (c) Ground truth. (d) and (e) Results of "Base" and "Base + BEM," respectively.

three datasets with 10% labeled data during training. The same hyperparameter settings are used for all the comparisons. In the subsequent trials, "Base" refers to the fundamental segmentation model devoid of BEM or SFLM. "Base + BEM" integrates the BEM into the "Base" model, and "Base + SFLM" refers to the model with SFLM.

Table V presents the results of ablation studies on the BESSL-Net segmentation model with or without BEM and SFLM. Generally, the empirical evidence attests to the superior performance of BESSL-Net compared with other models on the three datasets, achieving the highest IoUc and OA. The values of IoUc and OA decrease when either module is removed during training, highlighting the importance of both the modules in achieving optimal performance. Furthermore, the base segmentation model trained without either module has the lowest IoUc and OA values on LEVIR-CD and WHU-CD, proving further evidence of the effectiveness of BEM and SFLM. After analyzing the SYSU-CD, it is found that adding SFLM to the base segmentation model results in a minor decrease in IoUc and OA. Whereas the identification ability of the whole model is notably improved after introducing SFLM to "Base + BEM," showing that the SFLM is still an effective module for feature representation. Refining feature representation of boundary and changed areas information can significantly improve the extraction of changed areas, as evidenced by the significant impact of BEM and SFLM on IoUc, while only a moderate effect on OA. Furthermore, BEM and SFLM can notably boost the segmentation capacity of BESSL-Net for irregular multicategory objects, surpassing building datasets with more regular boundaries.

To provide a more perceptible demonstration of the effectiveness of the two submodules, we have visualized the results and feature maps for each submodule (Figs. 7 and 8). As depicted in Fig. 7(d) and (e), it is evident that "Base + BEM" exhibits improved boundary detection performance on the three datasets. Furthermore, BEM effectively mitigates omission and commission errors, leading to more accurate and reliable boundary detection results. In the second example from the WHU-CD, although the boundary detection performance has been significantly enhanced, an unidentified region remains in the lower right corner. This further emphasizes that while the BEM can greatly improve the model's ability to recognize boundaries, it overlooks fine-grained features, resulting in some errors. However, the SFLM can alleviate these issues, as illustrated in Fig. 8(c) and (f), where feature details are further emphasized. It is worth noting that when SFLM is added alone, there is a slight decrease in OA. Nevertheless, when both BEM and SFLM are combined, their complementary strengths come into play, significantly enhancing the model's recognition accuracy. This combination of modules further reinforces their effectiveness.

Overall, the addition of BEM and SFLM in tandem addresses the limitations of each module and brings about notable improvements in the model's boundary recognition and feature detailing capabilities. These findings from ablation studies emphasize the critical role of BEM and SFLM modules in boosting the identification ability of BESSL-Net.

2) Loss Functions: In this article, BESSL-Net is optimized using a combined loss of a supervised loss, a pseudolabeling loss, and a contrastive loss. To analyze different losses, we also carry out ablation experiments on optimization with various losses on the three datasets with 10% labeled images during training using the same hyperparameter settings. In subsequent trials, Θ_{sup}, Θ_{pseudo}, and Θ_{contra} represent the supervised loss, the pseudolabeling loss, and the contrastive loss, respectively.

The quantitative results of ablation studies with various losses are summarized in Table VI. Combining all the losses achieves the best performance on the three datasets, achieving the highest IoUc and OA. Compared with the optimization using only the supervised loss, incorporating the pseudolabeling loss into optimization leads to a considerable improvement in model performance, with IoUc rising by a minimum of 3%. However, it should be highlighted that the model optimized with both the supervised and contrastive losses simultaneously exhibits the worst performance, performing even worse than using the supervised loss only. While the best performance

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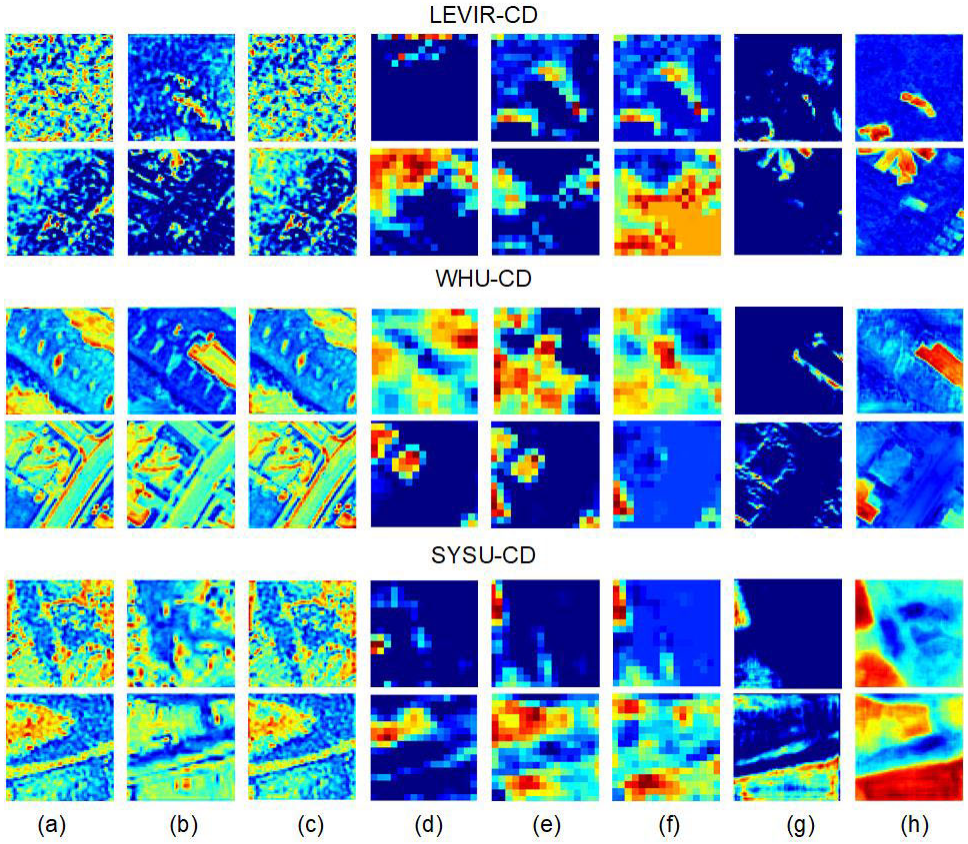


Fig. 8. Visualization results of intermediate feature maps. (a) and (b) Primitive features of the bitemporal image pairs. (c) Primitive features obtained from SFLM. (d) and (e) Semantic features of the bitemporal images. (f) Semantic features from SFLM. (g) Edge features from BEM. (h) Last change features.

TABLE VI
EFFECTIVENESS OF DIFFERENT LOSSES ON THE THREE DATASETS WITH 10% LABELED DATA IN TRAINING

Method	LEVIR-CD		WHU-CD		SYSU-CD	
	IoUc	OA	IoUc	OA	IoUc	OA
Θ_{sup}	0.7674	0.9864	0.693	0.9842	0.6141	0.8987
$\Theta_{sup}+\Theta_{pseudo}$	0.7946	0.9883	0.767	0.9894	0.6681	0.9141
$\Theta_{sup}+\Theta_{contra}$	0.7594	0.9858	0.686	0.9842	0.6104	0.8972
$\Theta_{sup}+\Theta_{pseudo}+\Theta_{contra}$	**0.8114**	**0.9895**	**0.7807**	**0.9899**	**0.6687**	**0.9143**

The best results are highlighted in bold.

TABLE VII
VALIDATION OF DATA PREPROCESSING ON THE THREE DATASETS WITH 10% LABELED DATA

Method	LEVIR-CD		WHU-CD		SYSU-CD	
	IoUc	OA	IoUc	OA	IoUc	OA
All	0.8076	0.9873	0.7121	0.9872	0.6486	0.9082
Non-unchange	**0.8114**	**0.9895**	**0.7807**	**0.9899**	**0.6687**	**0.9143**

The best results are highlighted in bold.

is observed when the CL loss and the pseudolabeling loss are introduced together into optimization, indicating that the contrastive loss is only effective when used in conjunction with pseudolabeling. In addition, the detection performance is substantially improved when optimized with all the three loss functions on the WHU-CD dataset, resulting in an almost 2% increase in IoUc.

3) Data Preprocessing: This article excludes all the images with only one classification (unchanged area) from the training set. More details of data preprocessing can be seen in Section IV-A2. The effectiveness of the preprocessing technique is evaluated using 10% labeled images in the testing sets of the three datasets. The same hyperparameter settings are used for both the original and preprocessed training sets. In the subsequent trials, "All" refers to the original training sets, which have 7120/4608/12,000 images in the training sets for LEVIR-CD, WHU-CD, and SYSU-CD, respectively. The training sets used for this article are referred to as "Non-unchange."

As shown in Table VII, the training sets preprocessed with selection outperform the original sets on all the three datasets with better IoUc and OA. Specifically, the results of "Non-unchange" can reach or even exceed the accuracy of "All" with considerably less training data, indicating that the data preprocessing technique successfully facilitates further feature learning of changed areas and enhances the overall detection performance. Moreover, it is essential to highlight that IoUc increases by 6% in the WHU-CD dataset, even though the image pairs in the training sets after selection are fewer than those in the validation and testing sets. This improvement provides compelling evidence of the efficacy of our data preprocessing technique.

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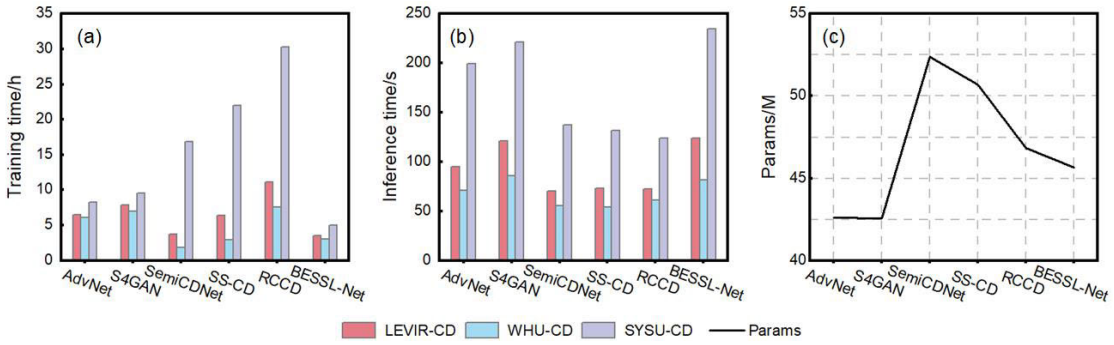


Fig. 9. Comparisons between various networks regarding time efficiency. (a) Training time. (b) Inference time. (c) Params.

TABLE VIII

EFFECTIVENESS OF DIFFERENT CROP SIZES ON THE LEVIR-CD AND WHU-CD DATASETS WITH 10% LABELED DATA IN TRAINING

Method	LEVIR-CD		WHU-CD	
	IoUc	OA	IoUc	OA
512*512	0.8033	0.9889	0.7000	0.9864
256*256	**0.8114**	**0.9895**	**0.7807**	**0.9899**

The best results are highlighted in bold.

We also investigate how the size of training slices affects the discrimination ability of BESSL-Net. For the LEVIR-CD and WHU-CD with large image sizes, each image is sliced into 512 × 512 and 256 × 256 patches with no overlap. The validation for different slice sizes is evaluated using 10% labeled images on the testing sets of both the datasets. Table VIII summarizes the quantitative results within IoUc and OA.

Theoretically, larger training slices can provide more context information and lead to higher accuracy [5]. However, it is found that training with larger patch sizes results in lower accuracy in both IoUc and OA. Specifically, on the WHU-CD dataset, the results with larger sizes are significantly worse than those with smaller sizes, with a 4% reduction in IoUc. Our BESSL-Net performs directly classwise feature representation learning with the feature bank and achieves good results with smaller patch images for training. In contrast, the rich spatial information in large images becomes a kind of noise for our model, negatively affecting the detection performance. In addition, using larger images requires twice as much time, as well as more capacity and memory, which can be impractical for some applications.

4) Model Efficiency: To comprehensively evaluate the model performance, the computational complexity and training time are reported in Fig. 9. The model efficiency of all the methods is assessed based on training time, inference time, and the number of model parameters (Params) on a computing server equipped with an NVIDIA RTX 3080Ti GPU. Training time is measured using the three datasets with 10% labeled images. As shown in Fig. 9, the relationship between the Params and training time is not necessarily linear. Comparing the segmentation models with the CD models, it is observed that the segmentation models have fewer parameters, whereas the training time of both the models is not minimal. Moreover, we have found that training time is highly affected by the size of the dataset, and it can vary significantly among different datasets in the other three CD models. The inference time of BESSL-Net is greater than that of other models, especially on a large dataset like SYSU-CD [Fig. 9(b)], which could be attributed to our teacher–student structure. Nevertheless, it is essential to note that our BESSL-Net delivers superior model quality with a relatively short training time, achieving optimal performance. Comparatively, our model exhibits the least overall running time with the fewest parameters among the four CD models.

V. CONCLUSION

Due to the limited availability of labeled data and the complexity of RS images, it is challenging to identify the changed areas and boundary pixels precisely. Therefore, this article puts forward a BESSL-Net on the basis of dual encoders and one decoder segmentation backbone for CD tasks. The BESSL-Net leverages consistency regularization, pseudolabeling, and CL to maximize the usage of unlabeled images. Furthermore, the BEM and the change SFLM are proposed between the dual encoders and the decoder to generate discriminative features of change areas with precise boundary information. Specifically, BEM is applied to extract accurate boundary pixels based on both the primitive and semantic features obtained from the encoders. At the same time, SFLM is used to generate discriminative change features using various combinations of multilevel features. On the three open-source challenging CD datasets, it is evident that the BESSL-Net outperforms SOTA semi-supervised methods on both the metrics, OA and IoUc. Moreover, introducing the BEM and SFLM in our BESSL-Net significantly enhances both the identification and boundary detection performance. However, it should be noted that while BESSL-Net performs satisfactorily in building datasets such as LEVIR-CD and WHU-CD, it has limitations on more complex objects and scenes such as the SYSU-CD dataset. Subsequent research will further boost the identification performance of changed areas, especially for irregular and complicated objects.

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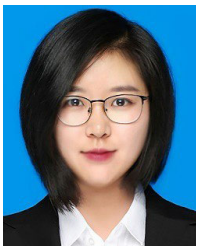
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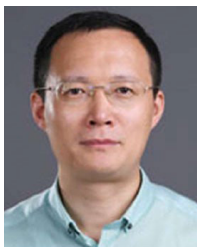
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